
```
1 .
2 . use merged_2017_1985.dta, clear

3 . xtset id year

    Panel variable: id (strongly balanced)
    Time variable: year, 2011 to 2023
               Delta: 1 unit

4 .
5 . * --- Rename to publication convention ---
6 . rename speedz      carb_dir

7 . rename speedq_raw  carb_lc

8 . rename speedq      carb_lc_w

9 . rename xpca        fin_green

10 . rename credit     fin_cred

11 . rename gdp        econ_gdp

12 . rename people     econ_pop

13 . rename fdi        econ_fdi
```

```
14 . rename environment reg_env
15 . rename famo reg_str
16 . rename lvdian ener_rnw
17 .
18 . * --- Generate derived variables ---
19 . gen ener_therm = ln(火电厂利用小时)
20 . label var ener_therm "Log thermal utilisation hours"
21 .
22 . * --- Label all variables ---
23 . label var carb_dir "Direct carbon intensity growth"
24 . label var carb_lc "Lifecycle carbon intensity growth"
25 . label var carb_lc_w "Lifecycle carbon intensity growth (winsorised)"
26 . label var fin_green "Green finance composite index"
27 . label var fin_cred "Green credit ratio"
28 . label var econ_gdp "Per capita GDP"
29 . label var econ_pop "Population"
30 . label var econ_fdi "Foreign direct investment"
31 . label var reg_env "Environmental regulatory investment"
32 . label var reg_str "Regulatory stringency"
```

```

33 . label var ener_rnw "Renewable electricity share"

34 .
35 . * --- Define control set ---
36 . global controls "econ_gdp econ_pop econ_fdi reg_env reg_str"

37 .
38 . save analysis_renamed.dta, replace
    file analysis_renamed.dta saved

39 .
40 .
41 . * =====
42 . * 1. BASELINE REGRESSIONS (Table 1 & Figure 1 Panel a)
43 . * =====
44 .
45 . display _n "=====

=====

46 . display " SECTION 1: BASELINE REGRESSIONS"
    SECTION 1: BASELINE REGRESSIONS

47 . display "=====

=====

48 .
49 . * — 1a. Lifecycle carbon intensity (carb_lc) —
50 .
51 . est clear

52 .
53 . * (1) RE: fin_green only
54 . xtreg carb_lc fin_green, re vce(cluster id)

Random-effects GLS regression              Number of obs    =      36
> 0
Group variable: id                        Number of groups   =      3
> 0

```

```

R-squared:                               Obs per group:
      Within = 0.0202                      min =      1
> 2                                     avg =     12.
      Between = 0.0060                      max =      1
> 0
      Overall = 0.0090
> 2

Wald chi2(1) =      4.9
> 9
corr(u_i, X) = 0 (assumed)                Prob > chi2 =     0.025
> 5

```

(Std. err. adjusted for 30 clusters in id)

```

> )

```

	Coefficient	Robust std. err.	z	P> z	[95% conf. interval	
carb_lc						
fin_green	.0173886	.0077853	2.23	0.026	.0021298	.032647
_cons	.0466938	.0315388	1.48	0.139	-.0151211	.108508
sigma_u	0					
sigma_e	.14026313					
rho	0	(fraction of variance due to u_i)				

```

> -

```

```

55 . est store lc_1

56 . estadd local ProvFE "No"

    added macro:
        e(ProvFE) : "No"

57 . estadd local YearFE "No"

    added macro:
        e(YearFE) : "No"

58 . estadd local Controls "No"

    added macro:
        e(Controls) : "No"

59 .
60 . * (2) Province + Year FE, no controls
61 . xtreg carb_lc fin_green i.year, fe vce(cluster id)

Fixed-effects (within) regression      Number of obs      =      36
> 0                                     Number of groups   =      3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.0697                      min =      1
> 2                                     avg =     12.
    Between = 0.0060                      max =      1
> 0                                     Overall = 0.0501
> 2

                                     F(12, 29)      =      4.6
> 9                                     Prob > F      =      0.000
corr(u_i, Xb) = -0.3938
> 3

```

$\geq)$

```
carb_lc | Coefficient std. err.      t    P>|t|    [95% conf. interval
> ]
```

 ≥ 2

year						
2013	-.0677285	.0478311	-1.42	0.167	-.1655541	.03009

> 7						
2014	-.0126235	.0344876	-0.37	0.717	-.0831585	.057911

> 5						
2015	-.0061252	.0248458	-0.25	0.807	-.0569406	.044690

> 1						
2016	-.0469993	.050254	-0.94	0.357	-.1497803	.055781

> 7						
2017	.0382109	.0371948	1.03	0.313	-.0378611	.114282

> 9						
2018	-.0190527	.042543	-0.45	0.658	-.1060629	.067957

> 5						
2019	-.0113888	.0561232	-0.20	0.841	-.1261736	.103395

> 9						
2020	.0339429	.0579891	0.59	0.563	-.0846582	.15254

> 4						
2021	.0035216	.0318756	0.11	0.913	-.0616713	.068714

> 6						
2022	.0186369	.0304884	0.61	0.546	-.0437188	.080992

> 6						
2023	.0250832	.037531	0.67	0.509	-.0516764	.101842

	11111	11111111	111111111	1111111111	11111111111	111111111111	1111111111111
> 7							

_cons	.1381422	.0872976	1.58	0.124	-.0404015	.316685
-------	----------	----------	------	-------	-----------	---------

sigma_u	.02721427	
sigma_e	.13902105	
rho	.03690632	(fraction of variance due to u_i)

 $\geq -$

```

62 . est store lc_2

63 . estadd local ProvFE "Yes"

    added macro:
        e(ProvFE) : "Yes"

64 . estadd local YearFE "Yes"

    added macro:
        e(YearFE) : "Yes"

65 . estadd local Controls "No"

    added macro:
        e(Controls) : "No"

66 .
67 . * (3) Province FE + controls, no year FE
68 . xtreg carb_lc fin_green $controls, fe vce(cluster id)

Fixed-effects (within) regression      Number of obs      =      36
> 0                                     Number of groups   =      3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.0441                      min =      1
> 2                                     avg =     12.
    Between = 0.0131                     max =      1
> 0                                     Overall = 0.0045
> 2

                                     F(6, 29)           =      2.2
> 4                                     Prob > F           =     0.066
corr(u_i, Xb) = -0.9542
> 9

```

```

                                (Std. err. adjusted for 30 clusters in id
> )
> -
      carb_lc | Coefficient   Robust      t    P>|t|    [95% conf. interval
> ]
-----|-----
> -
  fin_green |   .0321614   .0129673    2.48   0.019   .0056403   .058682
> 5
  econ_gdp |   .0588924   .0615982    0.96   0.347  -.0670901   .184874
> 8
  econ_pop |  -.1949445   .2218278   -0.88   0.387  -.6486332   .258744
> 2
  econ_fdi |   .0029253   .0055568    0.53   0.603  -.0084397   .014290
> 3
  reg_env |   1.637392   1.509763    1.08   0.287  -1.450421   4.72520
> 5
  reg_str |   .0776403   .3778146    0.21   0.839  -.6950773   .850357
> 9
    _cons |   1.035461   1.72166    0.60   0.552  -2.48573   4.55665
> 2
-----|-----
> -
  sigma_u |   .10655004
  sigma_e |   .13960723
  rho     |   .36808604   (fraction of variance due to u_i)
-----|-----
> -

```

```
69 . est store lc_3
```

```
70 . estadd local ProvFE "Yes"
```

```
added macro:
```

```
      e(ProvFE) : "Yes"
```


71 . estadd local YearFE "No"

added macro:

e(YearFE) : "No"

72 . estadd local Controls "Yes"

added macro:

e(Controls) : "Yes"

73 .

74 . * (4) Full model (BASELINE): two-way FE + controls

75 . xtreg carb_lc fin_green \$controls i.year, fe vce(cluster id)

Fixed-effects (within) regression Number of obs = 36

> 0

Group variable: id

Number of groups = 3

> 0

R-squared:

Obs per group:

Within = 0.0892

min = 1

> 2

Between = 0.0075

avg = 12.

> 0

Overall = 0.0026

max = 1

> 2

F(17, 29) = 3.5

> 9

corr(u_i, Xb) = -0.9887

Prob > F = 0.001

> 2

(Std. err. adjusted for 30 clusters in id

>)

> -

carb_lc	Coefficient	Robust std. err.	t	P> t	[95% conf. interval	
---------	-------------	---------------------	---	------	---------------------	--

>]

> -

fin_green	.0342365	.0155865	2.20	0.036	.0023586	.066114
-----------	----------	----------	------	-------	----------	---------

> 4

econ_gdp	.3082162	.2288853	1.35	0.189	-.1599067	.776339
----------	----------	----------	------	-------	-----------	---------

> 1

econ_pop	-.6772768	.4951268	-1.37	0.182	-1.689925	.335371
----------	-----------	----------	-------	-------	-----------	---------

```

> 2
econ_fdi | .0244634 .0129624 1.89 0.069 -.0020478 .050974
> 5
reg_env | 1.298327 2.060514 0.63 0.534 -2.915898 5.51255
> 2
reg_str | -.089468 .4882578 -0.18 0.856 -1.088067 .909131
> 4
      year
2013 | -.0966848 .0493244 -1.96 0.060 -.1975644 .004194
> 8
2014 | -.0632233 .0490174 -1.29 0.207 -.1634752 .037028
> 6
2015 | -.0795233 .0563369 -1.41 0.169 -.1947453 .035698
> 7
2016 | -.1400869 .0804836 -1.74 0.092 -.3046943 .024520
> 4
2017 | -.0933657 .0907524 -1.03 0.312 -.2789751 .092243
> 7
2018 | -.182013 .0979036 -1.86 0.073 -.3822484 .018222
> 3
2019 | -.202506 .1205183 -1.68 0.104 -.4489936 .043981
> 7
2020 | -.167615 .1196713 -1.40 0.172 -.4123703 .077140
> 4
2021 | -.2375045 .1600707 -1.48 0.149 -.5648859 .089876
> 9
2022 | -.3534331 .2101363 -1.68 0.103 -.7832101 .07634
> 4
2023 | -.3655003 .2105595 -1.74 0.093 -.7961428 .065142
> 2
      _cons | 2.440218 2.207539 1.11 0.278 -2.074705 6.95514
> 1
-----
> -
      sigma_u | .29209025
      sigma_e | .13864544
      rho      | .81612119 (fraction of variance due to u_i)
-----
> -

```

```

76 . est store lc_4

77 . estadd local ProvFE "Yes"

    added macro:
        e(ProvFE) : "Yes"

78 . estadd local YearFE "Yes"

    added macro:
        e(YearFE) : "Yes"

79 . estadd local Controls "Yes"

    added macro:
        e(Controls) : "Yes"

80 .
81 . esttab lc_1 lc_2 lc_3 lc_4 using "Table1_carb_lc.rtf", replace ///
    >      b(%9.4f) se(%9.4f) star(* 0.10 ** 0.05 *** 0.01) ///
    >      keep(fin_green) ///
    >      stats(N r2_within ProvFE YearFE Controls, ///
    >      labels("Observations" "Within R²" "Province FE" "Year FE" "Contro
    > ls")) ///
    >      title("Table 1a. Green finance → lifecycle carbon intensity growth (car
    > b_lc)")
    (output written to Table1_carb_lc.rtf)

82 .
83 . * — 1b. Direct carbon intensity (carb_dir) —
84 .
85 . est clear

86 .

```

```

87 . * (1) RE: fin_green only
88 . xtreg carb_dir fin_green, re vce(cluster id)

```

```

Random-effects GLS regression           Number of obs   =       36
> 0                                     Number of groups  =        3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.0284                      min =         1
> 2                                     avg  =       12.
    Between = 0.0016                     max =         1
> 0                                     Overall = 0.0131
> 2                                     Wald chi2(1)      =        5.7
                                     Prob > chi2        =       0.016
> 6
corr(u_i, X) = 0 (assumed)
> 4

```

(Std. err. adjusted for 30 clusters in id)

```

> )

```

	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
carb_dir						
fin_green	-.0131737	.0054891	-2.40	0.016	-.0239321	-.002415
_cons	-.0796651	.0235693	-3.38	0.001	-.12586	-.033470
sigma_u	0					
sigma_e	.08723827					
rho	0	(fraction of variance due to u_i)				

```

> -

```

```

89 . est store dir_1

90 . estadd local ProvFE "No"

    added macro:
        e(ProvFE) : "No"

91 . estadd local YearFE "No"

    added macro:
        e(YearFE) : "No"

92 . estadd local Controls "No"

    added macro:
        e(Controls) : "No"

93 .
94 . * (2) Province + Year FE, no controls
95 . xtreg carb_dir fin_green i.year, fe vce(cluster id)

Fixed-effects (within) regression      Number of obs      =      36
> 0                                     Number of groups   =      3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.1290                      min =      1
> 2                                     avg =     12.
    Between = 0.0016                     max =      1
> 0                                     F(12, 29)          =      3.8
    Overall = 0.1029                     Prob > F           =     0.001
> 2                                     corr(u_i, Xb) = -0.2763
> 4

```

$\geq)$

		Robust			
carb_dir	Coefficient	std. err.	t	P> t	[95% conf. interval
>]					

> 8

> 5

> 3

> 3

> 5

> 4

> 7

 ≥ 2

> 6

sigma_u		.02076218	
sigma_e		.08401664	
rho		.05755354	(fraction of variance due to u_i)

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```

96 . est store dir_2

97 . estadd local ProvFE "Yes"

    added macro:
          e(ProvFE) : "Yes"

98 . estadd local YearFE "Yes"

    added macro:
          e(YearFE) : "Yes"

99 . estadd local Controls "No"

    added macro:
          e(Controls) : "No"

100 .
101 . * (3) Province FE + controls, no year FE
102 . xtreg carb_dir fin_green $controls, fe vce(cluster id)

Fixed-effects (within) regression      Number of obs      =      36
> 0                                     Number of groups   =      3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.1546                      min =      1
> 2                                     avg =     12.
    Between = 0.0401                      max =      1
> 0                                     Overall = 0.0032
> 2

                                     F(6, 29)      =      3.9
> 6                                     Prob > F      =     0.005
corr(u_i, Xb) = -0.9671
> 2

```

```

                                (Std. err. adjusted for 30 clusters in id
> )
> -
      carb_dir | Coefficient   Robust      t    P>|t|    [95% conf. interval
> ]
-----+-----
> -
    fin_green |   -.0330631   .0107392   -3.08   0.005   -.0550273   -.011098
> 9
    econ_gdp |    .1288038   .0460365    2.80   0.009    .0346486    .22295
> 9
    econ_pop |    .0547248   .1728216    0.32   0.754   -.2987351    .408184
> 7
    econ_fdi |   -.0027491   .0033842   -0.81   0.423   -.0096705    .004172
> 4
    reg_env |    1.211725   1.065572    1.14   0.265   -.9676146    3.39106
> 5
    reg_str |   -.3291598   .2053783   -1.60   0.120   -.7492056    .09088
> 6
      _cons |   -1.877413   1.211342   -1.55   0.132   -4.354886    .600059
> 9
-----+-----
> -
      sigma_u |    .14449791
      sigma_e |    .08200008
      rho     |    .75640864   (fraction of variance due to u_i)
-----+-----
> -

```

```
103 . est store dir_3
```

```
104 . estadd local ProvFE "Yes"
```

```
added macro:
```

```
      e(ProvFE) : "Yes"
```


105 . estadd local YearFE "No"

added macro:

e(YearFE) : "No"

106 . estadd local Controls "Yes"

added macro:

e(Controls) : "Yes"

107 .

108 . * (4) Full model: two-way FE + controls

109 . xtreg carb_dir fin_green \$controls i.year, fe vce(cluster id)

```
Fixed-effects (within) regression          Number of obs   =       36
> 0                                         Number of groups =        3
Group variable: id
> 0

R-squared:                                Obs per group:
    Within = 0.1909                        min =         1
> 2                                         avg  =       12.
    Between = 0.0100                       max  =         1
> 0                                         Overall = 0.0072
> 2                                         F(17, 29)       =        5.5
> 4                                         Prob > F        =       0.000
corr(u_i, Xb) = -0.9672
```

(Std. err. adjusted for 30 clusters in id)

>)

	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
carb_dir						
fin_green	-.0305255	.0097058	-3.15	0.004	-.050376	-.010674
econ_gdp	.3968386	.1831329	2.17	0.039	.0222898	.771387
econ_pop	-.3702915	.3342421	-1.11	0.277	-1.053893	.313310

```

> 5 econ_fdi | .0061654 .009794 0.63 0.534 -.0138657 .026196
> 4 reg_env | .8840682 1.504554 0.59 0.561 -2.19309 3.96122
> 6 reg_str | -.1284798 .3101874 -0.41 0.682 -.7628842 .505924
> 7
    year
2013 | -.0344621 .0261064 -1.32 0.197 -.0878558 .018931
> 5 2014 | -.0686352 .0377583 -1.82 0.079 -.1458596 .008589
> 3 2015 | -.0653492 .0485193 -1.35 0.188 -.1645824 .03388
> 4 2016 | -.0834471 .0614402 -1.36 0.185 -.2091064 .042212
> 2 2017 | -.0963871 .0663132 -1.45 0.157 -.2320128 .039238
> 7 2018 | -.150507 .0745329 -2.02 0.053 -.3029439 .001929
> 9 2019 | -.1790881 .0935387 -1.91 0.065 -.3703962 .012219
> 9 2020 | -.1828487 .09745 -1.88 0.071 -.3821563 .01645
> 9 2021 | -.2089793 .1272812 -1.64 0.111 -.4692985 .0513
> 4 2022 | -.2802401 .1654642 -1.69 0.101 -.6186524 .058172
> 2 2023 | -.2768474 .1634912 -1.69 0.101 -.6112245 .057529
> 7
    _cons | -1.016177 1.129889 -0.90 0.376 -3.327059 1.29470
> 5
-----
> -
    sigma_u | .15678189
    sigma_e | .08161964
    rho      | .78677113 (fraction of variance due to u_i)
-----
> -

```

```

110 . est store dir_4

111 . estadd local ProvFE "Yes"

    added macro:
        e(ProvFE) : "Yes"

112 . estadd local YearFE "Yes"

    added macro:
        e(YearFE) : "Yes"

113 . estadd local Controls "Yes"

    added macro:
        e(Controls) : "Yes"

114 .
115 . esttab dir_1 dir_2 dir_3 dir_4 using "Table1_carb_dir.rtf", replace ///
    >      b(%9.4f) se(%9.4f) star(* 0.10 ** 0.05 *** 0.01) ///
    >      keep(fin_green) ///
    >      stats(N r2_within ProvFE YearFE Controls, ///
    >          labels("Observations" "Within R²" "Province FE" "Year FE" "Contro
    > ls")) ///
    >      title("Table 1b. Green finance → direct carbon intensity growth (carb_d
    > ir)")
    (output written to Table1_carb_dir.rtf)

116 .
117 .
118 . * =====
119 . * 3. HETEROGENEITY ANALYSIS (Table 3)
120 . * =====
121 .

```

```

122 . display _n "=====
=====

123 . display    " SECTION 3: HETEROGENEITY ANALYSIS"
    SECTION 3: HETEROGENEITY ANALYSIS

124 . display    "=====
=====

125 .
126 . * — 3a. By renewable energy endowment —
127 .
128 . bysort year: egen med_ener_rnw = median(ener_rnw)

129 . gen high_renewable = (ener_rnw >= med_ener_rnw)

130 .
131 . xtreg carb_dir fin_green $controls i.year if high_renewable == 1, fe

    Fixed-effects (within) regression           Number of obs   =       18
    > 0                                         Number of groups  =        1
    Group variable: id
    > 9

    R-squared:                                Obs per group:
        Within   = 0.3358                      min =
    > 1
        Between  = 0.0537                      avg  =       9.
    > 5
        Overall  = 0.0228                      max  =       1
    > 2

                                         F(17, 144)       =       4.2
    > 8
    corr(u_i, Xb) = -0.9756                  Prob > F         =       0.000
    > 0

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_dir						
> -	fin_green	-.0319657	.0151617	-2.11	0.037	-.0619339	-.001997
> 5	econ_gdp	.6476821	.1785955	3.63	0.000	.2946746	1.0006
> 9	econ_pop	-1.020558	.4039385	-2.53	0.013	-1.818972	-.222142
> 8	econ_fdi	.010432	.0185092	0.56	0.574	-.0261527	.047016
> 8	reg_env	3.435385	2.136004	1.61	0.110	-.7865876	7.65735
> 8	reg_str	-.2102918	.8301531	-0.25	0.800	-1.851152	1.43056
> 8							
	year						
	2013	-.0284163	.040417	-0.70	0.483	-.1083035	.051470
> 9	2014	-.1219093	.0502547	-2.43	0.017	-.2212414	-.022577
> 1	2015	-.1216925	.0559654	-2.17	0.031	-.2323123	-.011072
> 7	2016	-.1479815	.0686352	-2.16	0.033	-.2836441	-.012318
> 9	2017	-.1567592	.0834463	-1.88	0.062	-.321697	.008178
> 6	2018	-.2458926	.0975486	-2.52	0.013	-.4387046	-.053080
> 6	2019	-.304468	.1086046	-2.80	0.006	-.5191332	-.089802
> 8	2020	-.3144026	.1168796	-2.69	0.008	-.5454238	-.083381
> 4	2021	-.3000663	.1363035	-2.20	0.029	-.5694803	-.030652
> 2	2022	-.4559374	.1678078	-2.72	0.007	-.7876221	-.124252
> 6	2023	-.4475429	.1747872	-2.56	0.011	-.7930229	-.102062
> 9							
	_cons	1.867396	2.630953	0.71	0.479	-3.332879	7.06767
> 2							

```

> -
      sigma_u | .27670641
      sigma_e | .09713972
      rho     | .89028087   (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(18, 144) = 1.46                Prob > F = 0.113
> 5

```

132 . est store het_dir_hirnw

133 . xtreg carb_dir fin_green \$controls i.year if high_renewable == 0, fe

```

Fixed-effects (within) regression                Number of obs   =       18
> 0
Group variable: id                             Number of groups =        2
> 0

R-squared:                                     Obs per group:
      Within = 0.1234                                min =
> 1
      Between = 0.0000                                avg  =        9.
> 0
      Overall = 0.0001                                max  =        1
> 2

                                                F(17, 143)      =       1.1
> 8
corr(u_i, Xb) = -0.9844                      Prob > F         =       0.284
> 9

```

	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
carb_dir						
fin_green	-.0132153	.0102288	-1.29	0.198	-.0334345	.007003
econ_gdp	-.0125882	.1059106	-0.12	0.906	-.2219409	.196764
econ_pop	.2207587	.2610116	0.85	0.399	-.2951809	.736698
econ_fdi	-.0055141	.0133185	-0.41	0.679	-.0318407	.020812
reg_env	.4419678	.7873164	0.56	0.575	-1.114314	1.9982

```

> 5 reg_str | -.0626608 .4087188 -0.15 0.878 -.870572 .745250
> 4
    year
2013 | -.0395599 .0221424 -1.79 0.076 -.0833287 .004208
> 9
2014 | -.004853 .0249927 -0.19 0.846 -.0542558 .044549
> 8
2015 | -.0001298 .0290289 -0.00 0.996 -.0575109 .057251
> 4
2016 | .0060252 .0340545 0.18 0.860 -.06129 .073340
> 4
2017 | .0208541 .0434155 0.48 0.632 -.0649649 .106673
> 2
2018 | .0059887 .0523658 0.11 0.909 -.0975224 .109499
> 7
2019 | .0147577 .0597056 0.25 0.805 -.1032618 .132777
> 2
2020 | .0313777 .0650947 0.48 0.631 -.0972944 .160049
> 8
2021 | .0012807 .0795669 0.02 0.987 -.1559986 .158560
> 1
2022 | .0630329 .1262393 0.50 0.618 -.1865033 .312569
> 1
2023 | .066974 .1313056 0.51 0.611 -.1925767 .326524
> 8
    _cons | -1.700845 1.594955 -1.07 0.288 -4.853581 1.45189
> 1
-----
> -
    sigma_u | .15104301
    sigma_e | .05401689
    rho      | .88660634 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(19, 143) = 1.03 Prob > F = 0.429
> 2

```

134 . est store het_dir_lornw

135 . xtreg carb_lc fin_green \$controls i.year if high_renewable == 1, fe

```
Fixed-effects (within) regression      Number of obs   =      18
> 0                                     Number of groups =      1
Group variable: id
> 9

R-squared:                             Obs per group:
    Within = 0.1710                      min =
> 1                                     avg =      9.
    Between = 0.0759                     max =      1
> 5                                     Overall = 0.0069
> 2                                     F(17, 144)      =      1.7
> 5                                     Prob > F         =      0.040
corr(u_i, Xb) = -0.9939
> 7
```

	Coefficient	Std. err.	t	P> t	[95% conf. interval	
carb_lc						
fin_green	.0322393	.0211886	1.52	0.130	-.0096415	.074120
econ_gdp	.7026239	.2495885	2.82	0.006	.2092935	1.19595
econ_pop	-1.500216	.5645068	-2.66	0.009	-2.616006	-.384425
econ_fdi	.0409307	.0258667	1.58	0.116	-.0101967	.092058
reg_env	-.0560635	2.985081	-0.02	0.985	-5.9563	5.84417
reg_str	-.4430667	1.160145	-0.38	0.703	-2.73618	1.85004
year						
2013	-.0534837	.056483	-0.95	0.345	-.1651266	.058159
2014	-.1124901	.0702313	-1.60	0.111	-.2513076	.026327


```

> 8      2015 |   -.1697838   .0782121   -2.17   0.032   -.3243758   -.015191
> 3      2016 |   -.2052878   .0959182   -2.14   0.034   -.3948773   -.015698
> 9      2017 |   -.2014709   .1166167   -1.73   0.086   -.4319726   .029030
> 7      2018 |   -.3271829   .1363248   -2.40   0.018   -.596639   -.057726
> 8      2019 |    -.358393   .1517757   -2.36   0.020   -.6583891   -.058396
> 7      2020 |   -.3300345    .16334   -2.02   0.045   -.6528883   -.007180
> 5      2021 |   -.4501905   .1904851   -2.36   0.019   -.8266985   -.073682
> 1      2022 |   -.6898959   .2345126   -2.94   0.004   -1.153428   -.226364
> 9      2023 |   -.7091136   .2442663   -2.90   0.004   -1.191924   -.226302
> 6      _cons |    5.367343    3.676775    1.46   0.147   -1.900078    12.6347
> -
>      sigma_u |   .51824057
>      sigma_e |   .13575345
>      rho     |   .93578797   (fraction of variance due to u_i)
> -
> F test that all u_i=0: F(18, 144) = 0.99                      Prob > F = 0.473
> 4

```

```
136 . est store het_lc_hirnw
```

```
137 . xtreg carb_lc fin_green $controls i.year if high_renewable == 0, fe
```

```

Fixed-effects (within) regression          Number of obs   =       18
> 0                                         Number of groups =       2
Group variable: id
> 0

```

```

R-squared:                               Obs per group:
    Within = 0.1770                        min =
> 1
    Between = 0.1648                      avg =      9.
> 0
    Overall = 0.0300                      max =      1
> 2

                                           F(17, 143)      =      1.8
> 1
corr(u_i, Xb) = -0.9540                  Prob > F      =      0.032
> 1

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
>]							
> -	fin_green	.0626725	.0262538	2.39	0.018	.0107769	.114568
> 1	econ_gdp	-.4382614	.2718354	-1.61	0.109	-.9755963	.099073
> 4	econ_pop	.363162	.6699253	0.54	0.589	-.9610741	1.68739
> 8	econ_fdi	-.0188082	.034184	-0.55	0.583	-.0863794	.04876
> 3	reg_env	2.727636	2.020765	1.35	0.179	-1.266795	6.72206
> 6	reg_str	-.2848446	1.049038	-0.27	0.786	-2.358469	1.7887
> 8							
	year						
	2013	-.1370362	.0568319	-2.41	0.017	-.2493754	-.024697
> 1	2014	.006838	.0641474	0.11	0.915	-.1199616	.133637
> 6	2015	.0372072	.0745069	0.50	0.618	-.11007	.184484
> 4	2016	-.0185464	.0874058	-0.21	0.832	-.1913209	.15422
> 8	2017	.1153099	.1114323	1.03	0.303	-.1049575	.335577
> 3	2018	.0901044	.1344046	0.67	0.504	-.1755721	.355780
> 9	2019	.1102602	.1532432	0.72	0.473	-.1926545	.413174

```

> 8
      2020 | .1746328 .1670752 1.05 0.298 -.1556233 .50488
> 9
      2021 | .2205778 .2042204 1.08 0.282 -.1831032 .624258
> 7
      2022 | .3781264 .324012 1.17 0.245 -.2623455 1.01859
> 8
      2023 | .4006343 .3370154 1.19 0.236 -.2655414 1.0668
> 1
      _cons | 1.696407 4.093691 0.41 0.679 -6.39556 9.78837
> 5
-----
> -
      sigma_u | .17600531
      sigma_e | .13864242
      rho      | .61709407 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(19, 143) = 0.66 Prob > F = 0.848
> 8

```

```
138 . est store het_lc_lornw
```

```
139 .
```

```

140 . esttab het_dir_hirnw het_dir_lornw het_lc_hirnw het_lc_lornw, ///
>      mtitle("carb_dir Hi-Rnw" "carb_dir Lo-Rnw" "carb_lc Hi-Rnw" "carb_lc Lo
> -Rnw") ///
>      keep(fin_green) b(%9.4f) se(%9.4f) star(* 0.1 ** 0.05 *** 0.01) ///
>      title("Table 3a. Heterogeneity by renewable energy endowment")

```

Table 3a. Heterogeneity by renewable energy endowment

	(1) carb_dir H~w	(2) carb_dir L~w	(3) carb_lc Hi~w	(4) carb_lc Lo~w
fin_green	-0.0320** (0.0152)	-0.0132 (0.0102)	0.0322 (0.0212)	0.0627** (0.0263)
N	180	180	180	180

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

```

141 .
142 . * — 3b. By region (East vs Central-West) —
143 .
144 . gen east = ///
>     inlist(地区, "北京","天津","河北","上海","江苏","浙江") | ///
>     inlist(地区, "福建","山东","广东","海南","辽宁")

145 .
146 . xtreg carb_dir fin_green $controls i.year if east == 1, fe

```

```

Fixed-effects (within) regression          Number of obs   =       13
> 2                                         Number of groups =       1
Group variable: id
> 1

R-squared:                                Obs per group:
    Within = 0.2004                        min =       1
> 2                                         avg =      12.
    Between = 0.0042                      max =       1
> 0                                         Overall = 0.0035
> 2

                                         F(17, 104)       =       1.5
> 3                                         Prob > F         =     0.097
corr(u_i, Xb) = -0.9933
> 5

```

	Coefficient	Std. err.	t	P> t	[95% conf. interval
carb_dir					
fin_green	-0.0177015	.0099107	-1.79	0.077	[-0.0373548, .001951]
econ_gdp	.2133907	.1359003	1.57	0.119	[-.0561046, .482886]
econ_pop	.0768561	.325801	0.24	0.814	[-.5692196, .722931]
econ_fdi	-.0059638	.0087988	-0.68	0.499	[-.0234122, .011484]
reg_env	-.1753533	.7555401	-0.23	0.817	[-1.673618, 1.32291]
reg_str	.4409187	.4906081	0.90	0.371	[-.5319756, 1.41381]

```

      year |
> 6 2013 | -.0190735 .0229796 -0.83 0.408 -.064643 .02649
> 3 2014 | -.0362555 .0270877 -1.34 0.184 -.0899714 .017460
> 8 2015 | -.0412818 .0325302 -1.27 0.207 -.1057904 .023226
> 6 2016 | -.0880045 .0396074 -2.22 0.028 -.1665475 -.009461
> 7 2017 | -.0427827 .0500645 -0.85 0.395 -.1420624 .05649
> 2 2018 | -.053808 .0602098 -0.89 0.374 -.1732063 .065590
> 1 2019 | -.0973921 .0690366 -1.41 0.161 -.2342943 .039510
> 9 2020 | -.1008272 .071324 -1.41 0.160 -.2422653 .040610
> 7 2021 | -.1294249 .0859921 -1.51 0.135 -.2999504 .041100
> 5 2022 | -.1184227 .1033552 -1.15 0.255 -.32338 .086534
> 8 2023 | -.1441415 .1088594 -1.32 0.188 -.3600138 .071730
> 5 _cons | -2.81169 1.981324 -1.42 0.159 -6.74073 1.1173
> -
      sigma_u | .23397112
      sigma_e | .04846703
      rho     | .95885462 (fraction of variance due to u_i)
> -
F test that all u_i=0: F(10, 104) = 1.42 Prob > F = 0.179
> 8

```

147 . est store het_dir_east

148 . xtreg carb_dir fin_green \$controls i.year if east == 0, fe

```
Fixed-effects (within) regression      Number of obs   =      22
> 8                                     Number of groups =      1
Group variable: id
> 9

R-squared:                             Obs per group:
    Within = 0.2505                      min =      1
> 2                                     avg =     12.
    Between = 0.0394                     max =      1
> 0                                     Overall = 0.0150
> 2                                     F(17, 192)      =      3.7
> 8                                     Prob > F         =     0.000
corr(u_i, Xb) = -0.9489
```

	Coefficient	Std. err.	t	P> t	[95% conf. interval	
carb_dir						
fin_green	-.0374242	.0130263	-2.87	0.005	-.0631172	-.011731
econ_gdp	.3652078	.1470499	2.48	0.014	.0751672	.655248
econ_pop	-.2044598	.3682168	-0.56	0.579	-.9307292	.521809
econ_fdi	.0274277	.0197191	1.39	0.166	-.0114662	.066321
reg_env	1.249163	1.567	0.80	0.426	-1.841583	4.33990
reg_str	-.9109772	.6484996	-1.40	0.162	-2.190076	.368121
year						
2013	-.0375658	.0341178	-1.10	0.272	-.1048596	.029728
2014	-.0770641	.0412651	-1.87	0.063	-.1584553	.004327

```

> 9      2015 |   -.0656937   .0461731   -1.42   0.156   -.1567653   .025377
> 6      2016 |   -.0588761   .0546719   -1.08   0.283   -.1667109   .048958
> 5      2017 |   -.0990008   .0684042   -1.45   0.149   -.2339211   .035919
> 1      2018 |   -.1694574   .0815031   -2.08   0.039   -.3302137   -.00870
> 7      2019 |   -.1830489   .0917144   -2.00   0.047   -.3639461   -.002151
> 9      2020 |   -.1826936   .0999842   -1.83   0.069   -.379902   .014514
> 8      2021 |   -.1871661   .1190942   -1.57   0.118   -.422067   .047734
> 2      2022 |    -.370141   .1804612   -2.05   0.042   -.7260821    -.014
> 5      2023 |   -.3500261   .1881897   -1.86   0.064   -.7212107   .021158
> 6      _cons |   -2.215539   2.149821   -1.03   0.304   -6.455837   2.0247
> -
>      sigma_u |   .16480461
>      sigma_e |   .09416412
>      rho     |   .75388536   (fraction of variance due to u_i)
> -
> F test that all u_i=0: F(18, 192) = 1.07                      Prob > F = 0.387
> 5

```

```
149 . est store het_dir_west
```

```
150 . xtreg carb_lc fin_green $controls i.year if east == 1, fe
```

```

Fixed-effects (within) regression          Number of obs   =       13
> 2
Group variable: id                        Number of groups  =       1
> 1

```

```

R-squared:                               Obs per group:
  Within = 0.1619                        min =      1
> 2                                     avg =     12.
> 0                                     max =      1
  Overall = 0.0013

                                     F(17, 104)      =      1.1
> 8                                     Prob > F       =      0.292
corr(u_i, Xb) = -0.9947
> 3

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
>]							
> -	fin_green	.0748348	.0320101	2.34	0.021	.0113576	.13831
> 2	econ_gdp	-.4693915	.4389381	-1.07	0.287	-1.339822	.401039
> 3	econ_pop	1.144015	1.05229	1.09	0.279	-.9427149	3.23074
> 5	econ_fdi	-.0107886	.0284189	-0.38	0.705	-.0671443	.045567
> 1	reg_env	2.811523	2.440284	1.15	0.252	-2.027652	7.65069
> 8	reg_str	.9511259	1.584593	0.60	0.550	-2.191181	4.09343
> 2							
	year						
> 5	2013	-.1007914	.0742209	-1.36	0.177	-.2479742	.046391
> 5	2014	-.0420828	.0874892	-0.48	0.632	-.2155772	.131411
> 5	2015	.0021971	.1050678	0.02	0.983	-.2061563	.210550
> 2	2016	-.0791659	.127926	-0.62	0.537	-.332848	.174516
> 5	2017	.0832605	.1617009	0.51	0.608	-.2373984	.403919
> 9	2018	.0228551	.1944687	0.12	0.907	-.3627837	.408493
	2019	-.0330586	.2229782	-0.15	0.882	-.4752327	.409115


```

> 5      2020 | .0714122 .230366 0.31 0.757 -.3854122 .528236
> 7      2021 | .1938932 .2777419 0.70 0.487 -.3568794 .744665
> 8      2022 | .2569904 .3338223 0.77 0.443 -.4049916 .918972
> 5      2023 | .2642899 .3516 0.75 0.454 -.4329462 .96152
> 6
      _cons | -4.390221 6.399387 -0.69 0.494 -17.08045 8.30000
> 4
-----
> -
      sigma_u | .64168828
      sigma_e | .15654145
      rho     | .94383002 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(10, 104) = 0.80          Prob > F = 0.631
> 4

```

```
151 . est store het_lc_east
```

```
152 . xtreg carb_lc fin_green $controls i.year if east == 0, fe
```

```

Fixed-effects (within) regression          Number of obs   =       22
> 8
Group variable: id                        Number of groups  =        1
> 9

R-squared:                                Obs per group:
      Within = 0.1576                                min =        1
> 2
      Between = 0.0260                                avg  =       12.
> 0
      Overall = 0.0249                                max  =        1
> 2

                                                F(17, 192)       =       2.1
> 1
corr(u_i, Xb) = -0.9291                    Prob > F          =       0.007
> 9

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
carb_lc							
fin_green		.0334344	.0173017	1.93	0.055	-.0006913	.067560
econ_gdp		.3401424	.195313	1.74	0.083	-.0450923	.725377
econ_pop		-.5369342	.4890689	-1.10	0.274	-1.501572	.427703
econ_fdi		.0575271	.0261911	2.20	0.029	.0058678	.109186
reg_env		.0641877	2.081304	0.03	0.975	-4.04097	4.16934
reg_str		-1.028218	.8613432	-1.19	0.234	-2.727129	.670691
year							
2013		-.0840333	.0453156	-1.85	0.065	-.1734136	.00534
2014		-.0543095	.0548087	-0.99	0.323	-.162414	.053795
2015		-.0927233	.0613275	-1.51	0.132	-.2136855	.028238
2016		-.1182189	.0726158	-1.63	0.105	-.261446	.025008
2017		-.1182711	.0908552	-1.30	0.195	-.2974735	.060931
2018		-.19388	.1082531	-1.79	0.075	-.4073981	.019638
2019		-.1821924	.1218159	-1.50	0.136	-.4224617	.058076
2020		-.1641904	.1327999	-1.24	0.218	-.4261245	.097743
2021		-.2786121	.158182	-1.76	0.080	-.5906096	.033385
2022		-.5442398	.2396902	-2.27	0.024	-1.017004	-.071475
2023		-.5510457	.2499552	-2.20	0.029	-1.044057	-.058034
_cons		.7665177	2.855412	0.27	0.789	-4.865486	6.39852

```

> -
      sigma_u | .13429403
      sigma_e | .12506965
      rho     | .53552047 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(18, 192) = 0.76          Prob > F = 0.750
> 0

```

```
153 . est store het_lc_west
```

```
154 .
```

```

155 . esttab het_dir_east het_dir_west het_lc_east het_lc_west, ///
>      mtitle("carb_dir East" "carb_dir C-W" "carb_lc East" "carb_lc C-W") ///
>      keep(fin_green) b(%9.4f) se(%9.4f) star(* 0.1 ** 0.05 *** 0.01) ///
>      title("Table 3b. Heterogeneity by region")

```

Table 3b. Heterogeneity by region

	(1) carb_dir East	(2) carb_dir C-W	(3) carb_lc East	(4) carb_lc C-W
fin_green	-0.0177* (0.0099)	-0.0374*** (0.0130)	0.0748** (0.0320)	0.0334* (0.0173)
N	132	228	132	228

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

```
156 .
```

```
157 . * — 3c. By economic development (GDP) —
```

```
158 .
```

```
159 . bysort year: egen med_gdp =median(econ_gdp)
```

```
160 . gen high_gdp = (econ_gdp >= med_gdp)
```

```
161 .
```

```
162 . xtreg carb_dir fin_green $controls i.year if high_gdp == 1, fe
```

```
Fixed-effects (within) regression      Number of obs   =      18
> 0                                     Number of groups =      1
Group variable: id
> 6

R-squared:                             Obs per group:
    Within = 0.2497                      min =
> 4                                     avg =     11.
    Between = 0.0103                     max =      1
> 2                                     Overall = 0.0083
> 2                                     F(17, 147)      =      2.8
> 8                                     Prob > F         =     0.000
corr(u_i, Xb) = -0.9712
```

```
> -
carb_dir | Coefficient Std. err.      t    P>|t|     [95% conf. interval
> ]
-----+-----
> -
fin_green | -.0355222   .0117903    -3.01   0.003    -.0588226   -.012221
> 8
econ_gdp  |  .4225818   .1473404     2.87   0.005    .1314027    .713760
> 9
econ_pop  | -.4857987   .3710584    -1.31   0.193    -1.219097    .247499
> 3
econ_fdi  | -.0032716   .0203462    -0.16   0.872    -.0434804    .036937
> 3
reg_env   |  .0416935   1.198034     0.03   0.972    -2.325902    2.40928
> 9
reg_str   |  .8333456   .7702204     1.08   0.281    -.6887896    2.35548
> 1
year
2013      | -.0289346   .0288698    -1.00   0.318    -.0859881    .02811
> 9
2014      | -.0499665   .0359322    -1.39   0.166    -.120977     .02104
```

```

> 4
2015 | -.0654571 .0437536 -1.50 0.137 -.1519245 .021010
> 3
2016 | -.1124562 .053454 -2.10 0.037 -.2180938 -.006818
> 6
2017 | -.0956074 .0678232 -1.41 0.161 -.2296419 .038427
> 1
2018 | -.1411593 .0815471 -1.73 0.086 -.3023153 .019996
> 7
2019 | -.1750228 .0925451 -1.89 0.061 -.3579135 .00786
> 8
2020 | -.1981579 .0981262 -2.02 0.045 -.3920781 -.004237
> 8
2021 | -.2383636 .1141621 -2.09 0.039 -.4639746 -.012752
> 6
2022 | -.2386984 .175466 -1.36 0.176 -.58546 .108063
> 3
2023 | -.2288086 .1825301 -1.25 0.212 -.5895306 .131913
> 4
_cons | -.2900208 3.076894 -0.09 0.925 -6.370681 5.79063
> 9
-----
> -
sigma_u | .16201453
sigma_e | .06946456
rho | .84471521 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(15, 147) = 1.22 Prob > F = 0.264
> 5

```

163 . est store het_dir_higdp

```
164 . xtreg carb_dir fin_green $controls i.year if high_gdp == 0, fe
```

```
Fixed-effects (within) regression      Number of obs   =      18
> 0                                     Number of groups =      1
Group variable: id
> 6

R-squared:                             Obs per group:
    Within = 0.2211                      min =
> 4                                     avg =      11.
    Between = 0.0006                     max =      1
> 2                                     Overall = 0.0387
> 2                                     F(17, 147)      =      2.4
> 6                                     Prob > F         =      0.002
corr(u_i, Xb) = -0.9042
> 1
```

	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
carb_dir						
fin_green	-.0294213	.0145353	-2.02	0.045	-.0581464	-.000696
econ_gdp	.3943604	.1513819	2.61	0.010	.0951944	.693526
econ_pop	-.3585122	.3749539	-0.96	0.341	-1.099509	.382484
econ_fdi	.0080739	.0147266	0.55	0.584	-.0210292	.03717
reg_env	2.190014	1.344176	1.63	0.105	-.4663911	4.84641
reg_str	-.5289065	.617087	-0.86	0.393	-1.748414	.690601
year						
2013	-.0368316	.0369829	-1.00	0.321	-.1099184	.036255
2014	-.0814313	.0427347	-1.91	0.059	-.1658851	.003022
2015	-.0611304	.0468086	-1.31	0.194	-.1536351	.031374

```

    2016 | -.0471926 .0546505 -0.86 0.389 -.1551948 .060809
> 6
    2017 | -.0902982 .0673171 -1.34 0.182 -.2233325 .04273
> 6
    2018 | -.1536859 .0802708 -1.91 0.057 -.3123197 .004947
> 8
    2019 | -.1787456 .0903653 -1.98 0.050 -.3573284 -.000162
> 7
    2020 | -.152534 .0996209 -1.53 0.128 -.3494081 .044340
> 1
    2021 | -.1709007 .1199745 -1.42 0.156 -.4079983 .066196
> 8
    2022 | -.2828492 .1654361 -1.71 0.089 -.6097895 .044091
> 2
    2023 | -.2871608 .1723198 -1.67 0.098 -.627705 .053383
> 4
    _cons | -1.037808 2.013718 -0.52 0.607 -5.017384 2.94176
> 9
-----
> -
    sigma_u | .10329059
    sigma_e | .09286283
    rho      | .55301143 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(15, 147) = 1.09 Prob > F = 0.374
> 5

```

```
165 . est store het_dir_logdp
```

```
166 . xtreg carb_lc fin_green $controls i.year if high_gdp == 1, fe
```

```

Fixed-effects (within) regression      Number of obs   =       18
> 0
Group variable: id                    Number of groups =       1
> 6

R-squared:                            Obs per group:
    Within = 0.1202                      min =
> 4
    Between = 0.1257                      avg =      11.
> 2
    Overall = 0.0000                      max =       1
> 2

```

```

> 8
corr(u_i, Xb) = -0.9941
> 1

```

F(17, 147) = 1.1
Prob > F = 0.286

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
>]							
> -	fin_green	.0500874	.0270759	1.85	0.066	-.0034208	.103595
> 6	econ_gdp	.3057971	.3383601	0.90	0.368	-.3628813	.974475
> 6	econ_pop	-1.189471	.8521173	-1.40	0.165	-2.873454	.494511
> 4	econ_fdi	.0117366	.0467241	0.25	0.802	-.080601	.104074
> 3	reg_env	3.552785	2.751227	1.29	0.199	-1.884281	8.98985
> 1	reg_str	-.0775206	1.768773	-0.04	0.965	-3.573029	3.41798
> 8							
	year						
	2013	-.1420062	.0662981	-2.14	0.034	-.2730268	-.010985
> 6	2014	-.0976531	.0825166	-1.18	0.239	-.2607252	.06541
> 9	2015	-.1008476	.100478	-1.00	0.317	-.2994156	.097720
> 5	2016	-.2226845	.1227545	-1.81	0.072	-.465276	.01990
> 7	2017	-.1232269	.1557527	-0.79	0.430	-.4310305	.184576
> 8	2018	-.2286481	.1872688	-1.22	0.224	-.5987349	.141438
> 7	2019	-.2699627	.2125253	-1.27	0.206	-.6899623	.150036
> 9	2020	-.2166828	.2253419	-0.96	0.338	-.662011	.228645
> 3	2021	-.2137999	.2621677	-0.82	0.416	-.7319044	.304304
> 6	2022	-.2683538	.4029489	-0.67	0.506	-1.064675	.527967
> 3	2023	-.2749843	.4191713	-0.66	0.513	-1.103365	.55339


```

> 6
      _cons |      7.179672   7.065935      1.02   0.311   -6.784264   21.1436
> 1
-----
> -
      sigma_u |      .51324704
      sigma_e |      .15952193
      rho     |      .91190755   (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(15, 147) = 0.59                      Prob > F = 0.876
> 4

```

```
167 . est store het_lc_higdp
```

```
168 . xtreg carb_lc fin_green $controls i.year if high_gdp == 0, fe
```

```

Fixed-effects (within) regression      Number of obs   =      18
> 0
Group variable: id                    Number of groups  =       1
> 6

R-squared:                            Obs per group:
      Within = 0.1718                      min =
> 4
      Between = 0.1512                      avg =      11.
> 2
      Overall = 0.0169                      max =       1
> 2

                                           F(17, 147)      =      1.7
> 9
corr(u_i, Xb) = -0.9802                  Prob > F         =      0.033
> 7

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
> -	fin_green	.0301043	.0180226	1.67	0.097	-.0055126	.065721
> 1	econ_gdp	.3787263	.1877015	2.02	0.045	.0077844	.749668
> 1	econ_pop	-.6813295	.4649128	-1.47	0.145	-1.600106	.237446
> 6	econ_fdi	.0247397	.0182597	1.35	0.178	-.0113458	.060825
> 2	reg_env	1.47835	1.66667	0.89	0.377	-1.815379	4.7720
> 8	reg_str	-.0556762	.7651383	-0.07	0.942	-1.567768	1.45641
> 6							
	year						
	2013	-.0596274	.0458558	-1.30	0.196	-.1502492	.030994
> 4	2014	-.0352018	.0529877	-0.66	0.508	-.1399178	.069514
> 2	2015	-.0714396	.0580389	-1.23	0.220	-.186138	.043258
> 8	2016	-.0690637	.0677623	-1.02	0.310	-.2029777	.064850
> 4	2017	-.0848514	.0834678	-1.02	0.311	-.2498032	.080100
> 4	2018	-.1622595	.0995293	-1.63	0.105	-.3589527	.034433
> 6	2019	-.170859	.1120457	-1.52	0.129	-.3922874	.050569
> 4	2020	-.1221998	.1235219	-0.99	0.324	-.3663079	.121908
> 3	2021	-.2585827	.1487587	-1.74	0.084	-.5525647	.035399
> 2	2022	-.3807937	.2051275	-1.86	0.065	-.7861735	.02458
> 6	2023	-.3969021	.2136628	-1.86	0.065	-.8191496	.025345
> 4							
	_cons	1.687359	2.496849	0.68	0.500	-3.246997	6.62171
> 5							

```

> -
      sigma_u | .24538726
      sigma_e | .11514245
      rho     | .81955494   (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(15, 147) = 0.82                Prob > F = 0.656
> 0

```

```
169 . est store het_lc_logdp
```

```
170 .
```

```

171 . esttab het_dir_higdp het_dir_logdp het_lc_higdp het_lc_logdp, ///
>      mtitle("carb_dir Hi-GDP" "carb_dir Lo-GDP" "carb_lc Hi-GDP" "carb_lc Lo
> -GDP") ///
>      keep(fin_green) b(%9.4f) se(%9.4f) star(* 0.1 ** 0.05 *** 0.01) ///
>      title("Table 3c. Heterogeneity by economic development")

```

Table 3c. Heterogeneity by economic development

	(1) carb_dir H~P	(2) carb_dir L~P	(3) carb_lc Hi~P	(4) carb_lc Lo~P
fin_green	-0.0355*** (0.0118)	-0.0294** (0.0145)	0.0501* (0.0271)	0.0301* (0.0180)
N	180	180	180	180

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

```
172 .
```

```
173 .
```

```
174 . * =====
```

```

175 . * 4. MECHANISM IDENTIFICATION: Control-and-Exclusion
176 . *      (Table 4 & Figure 2)
177 . * =====
178 .
179 . display _n "=====

=====

180 . display      " SECTION 4: MECHANISM IDENTIFICATION"
      SECTION 4: MECHANISM IDENTIFICATION

181 . display      "=====

=====

182 .
183 . * — Channel A: Thermal utilisation hours (ener_therm) —
184 .
185 . * Step 1: fin_green → ener_therm
186 . xtreg ener_therm fin_green $controls i.year, fe

Fixed-effects (within) regression          Number of obs      =      39
> 0
Group variable: id                        Number of groups   =       3
> 0

R-squared:                                Obs per group:
    Within = 0.4409                                min =       1
> 3
    Between = 0.0240                                avg  =     13.
> 0
    Overall = 0.0884                                max  =       1
> 3

                                                F(18, 342)          =     14.9
> 9
corr(u_i, Xb) = -0.7420                    Prob > F              =     0.000
> 0

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	ener_therm						
>]							
> -	fin_green	-.0326788	.0121587	-2.69	0.008	-.056594	-.008763
> 6	econ_gdp	.0745007	.1169211	0.64	0.524	-.1554742	.304475
> 7	econ_pop	-.3459226	.2856302	-1.21	0.227	-.9077358	.215890
> 5	econ_fdi	.0002948	.0144381	0.02	0.984	-.0281037	.028693
> 4	reg_env	3.100881	1.164659	2.66	0.008	.8100853	5.39167
> 8	reg_str	.1198191	.5896079	0.20	0.839	-1.039895	1.27953
> 3							
	year						
	2012	-.0844362	.0320396	-2.64	0.009	-.1474556	-.021416
> 8	2013	-.0723089	.0371738	-1.95	0.053	-.145427	.000809
> 2	2014	-.1283866	.0428519	-3.00	0.003	-.212673	-.044100
> 2	2015	-.2519192	.0471438	-5.34	0.000	-.3446474	-.159190
> 9	2016	-.3236358	.0538059	-6.01	0.000	-.4294679	-.217803
> 7	2017	-.3311483	.063487	-5.22	0.000	-.4560225	-.206274
> 1	2018	-.2783201	.0733724	-3.79	0.000	-.4226381	-.134002
> 1	2019	-.3038549	.0808923	-3.76	0.000	-.4629639	-.144745
> 9	2020	-.3052536	.0858389	-3.56	0.000	-.4740922	-.136414
> 9	2021	-.2444336	.0994698	-2.46	0.014	-.4400833	-.048783
> 9	2022	-.2670338	.1350704	-1.98	0.049	-.532707	-.001360
> 6	2023	-.2390382	.1404713	-1.70	0.090	-.5153346	.037258
> 3							
	_cons	10.45741	1.827967	5.72	0.000	6.861939	14.0528

```

> 9
-----
> -
      sigma_u      .23641722
      sigma_e      .11551045
      rho          .80728654   (fraction of variance due to u_i)
-----

> -
F test that all u_i=0: F(29, 342) = 17.47          Prob > F = 0.000
> 0

```

```
187 . est store mechA_s1
```

```
188 .
```

```
189 . * Step 2: fin_green → carb_dir | ener_therm
```

```
190 . xtreg carb_dir fin_green ener_therm $controls i.year, fe
```

```

Fixed-effects (within) regression          Number of obs   =       36
> 0
Group variable: id                        Number of groups =       3
> 0

R-squared:                                Obs per group:
      Within = 0.2582                               min =       1
> 2
      Between = 0.0024                               avg =      12.
> 0
      Overall = 0.0150                               max =       1
> 2

                                                F(18, 312)      =       6.0
> 3
corr(u_i, Xb) = -0.9612                    Prob > F        =     0.000
> 0

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_dir						
> -	fin_green	-.0236509	.0086764	-2.73	0.007	-.0407226	-.006579
> 2	ener_therm	.2046628	.0384555	5.32	0.000	.128998	.280327
> 7	econ_gdp	.3551007	.0936598	3.79	0.000	.170816	.539385
> 4	econ_pop	-.2686028	.2296718	-1.17	0.243	-.7205043	.183298
> 7	econ_fdi	.0064307	.0101314	0.63	0.526	-.0135039	.026365
> 3	reg_env	.2992445	.8481358	0.35	0.724	-1.369545	1.96803
> 4	reg_str	-.1601538	.4114025	-0.39	0.697	-.969628	.649320
> 3							
	year						
	2013	-.0345993	.0219303	-1.58	0.116	-.0777493	.008550
> 6	2014	-.0554304	.025516	-2.17	0.031	-.1056355	-.005225
> 2	2015	-.0256987	.029385	-0.87	0.382	-.0835166	.032119
> 2	2016	-.0272345	.0349789	-0.78	0.437	-.0960588	.041589
> 9	2017	-.0362475	.0424093	-0.85	0.393	-.1196919	.047196
> 9	2018	-.0988241	.0496225	-1.99	0.047	-.1964612	-.001187
> 1	2019	-.1205362	.0556385	-2.17	0.031	-.2300103	-.011062
> 1	2020	-.1230033	.0596996	-2.06	0.040	-.2404679	-.005538
> 6	2021	-.1582525	.0703886	-2.25	0.025	-.2967489	-.019756
> 1	2022	-.2246257	.0975589	-2.30	0.022	-.4165823	-.03266
> 9	2023	-.2260142	.1017513	-2.22	0.027	-.4262197	-.025808
> 7							
	_cons	-3.144217	1.484431	-2.12	0.035	-6.064977	-.223456

```

> 7
-----
> -
      sigma_u      .16325815
      sigma_e      .07827449
      rho          .81309109   (fraction of variance due to u_i)
-----

> -
F test that all u_i=0: F(29, 312) = 1.56                Prob > F = 0.035
> 8

```

```
191 . est store mechA_s2
```

```
192 .
```

```
193 . * Step 3: fin_green → carb_lc | ener_therm
```

```
194 . xtreg carb_lc fin_green ener_therm $controls i.year, fe
```

```

Fixed-effects (within) regression      Number of obs   =       36
> 0
Group variable: id                    Number of groups =        3
> 0

R-squared:                            Obs per group:
      Within = 0.1278                      min =         1
> 2
      Between = 0.0227                      avg  =       12.
> 0
      Overall = 0.0076                      max  =         1
> 2

                                          F(18, 312)      =        2.5
> 4
corr(u_i, Xb) = -0.9783                 Prob > F         =       0.000
> 6

```


		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
> -	fin_green	.0425683	.0150633	2.83	0.005	.0129299	.072206
> 8	ener_therm	.2480479	.0667633	3.72	0.000	.1166847	.37941
> 1	econ_gdp	.2576306	.1626046	1.58	0.114	-.0623097	.577570
> 8	econ_pop	-.5540319	.3987376	-1.39	0.166	-1.338587	.230522
> 8	econ_fdi	.024785	.0175894	1.41	0.160	-.0098239	.059393
> 8	reg_env	.5895306	1.472465	0.40	0.689	-2.307686	3.48674
> 7	reg_str	-.1278564	.7142437	-0.18	0.858	-1.5332	1.27748
> 7							
	year						
	2013	-.0968511	.0380736	-2.54	0.011	-.1717645	-.021937
> 6	2014	-.0472193	.0442988	-1.07	0.287	-.1343815	.039942
> 8	2015	-.0314676	.0510159	-0.62	0.538	-.1318463	.068911
> 1	2016	-.0719582	.0607275	-1.18	0.237	-.1914455	.04752
> 9	2017	-.0204775	.0736276	-0.28	0.781	-.1653469	.124391
> 9	2018	-.1193743	.0861506	-1.39	0.167	-.2888838	.050135
> 2	2019	-.131542	.096595	-1.36	0.174	-.321602	.05851
> 8	2020	-.0950833	.1036455	-0.92	0.360	-.2990159	.108849
> 2	2021	-.1760245	.122203	-1.44	0.151	-.4164707	.064421
> 7	2022	-.2860293	.1693739	-1.69	0.092	-.6192888	.047230
> 2	2023	-.3038914	.1766523	-1.72	0.086	-.6514719	.043689
> 2							
	_cons	-.1389295	2.577148	-0.05	0.957	-5.209717	4.93185

```
> 8
> -
```

sigma_u	.24361854	
sigma_e	.13589383	
rho	.76268549	(fraction of variance due to u_i)

```
> -
F test that all u_i=0: F(29, 312) = 0.63 Prob > F = 0.936
> 0
```

```
195 . est store mechA_s3
```

```
196 .
197 . esttab mechA_s1 mechA_s2 mechA_s3, ///
> mtitle("fin_green→therm" "carb_dir(+therm)" "carb_lc(+therm)") ///
> keep(fin_green ener_therm) b(%9.4f) se(%9.4f) star(* 0.1 ** 0.05 *** 0.
> 01) ///
> title("Table 4a. Channel A: thermal utilisation hours")
```

Table 4a. Channel A: thermal utilisation hours

	(1)	(2)	(3)
	fin_green→m	carb_dir(+~)	carb_lc(+t~)
fin_green	-0.0327*** (0.0122)	-0.0237*** (0.0087)	0.0426*** (0.0151)
ener_therm		0.2047*** (0.0385)	0.2480*** (0.0668)
N	390	360	360

Standard errors in parentheses

* p<0.1, ** p<0.05, *** p<0.01

```
198 .
199 . * — Channel B: Renewable electricity share (ener_rnw) —
200 .
201 . * Step 1: fin_green → ener_rnw
202 . xtreg ener_rnw fin_green $controls i.year, fe
```

```
Fixed-effects (within) regression      Number of obs   =      39
> 0
Group variable: id                    Number of groups =      3
> 0

R-squared:                            Obs per group:
    Within = 0.5043                      min =      1
> 3
    Between = 0.0322                     avg  =     13.
> 0
    Overall = 0.0511                     max  =      1
> 3

                                          F(18, 342)      =     19.3
> 3
corr(u_i, Xb) = -0.3364                 Prob > F        =     0.000
> 0
```

	Coefficient	Std. err.	t	P> t	[95% conf. interval	
ener_rnw						
fin_green	.016136	.0049654	3.25	0.001	.0063695	.025902
econ_gdp	-.1606313	.0477484	-3.36	0.001	-.2545488	-.066713
econ_pop	-.0050805	.1166461	-0.04	0.965	-.2345146	.224353
econ_fdi	.010313	.0058962	1.75	0.081	-.0012844	.021910
reg_env	-.6440346	.4756252	-1.35	0.177	-1.579554	.291484
reg_str	-.119374	.240785	-0.50	0.620	-.5929799	.35423
year						
2012	.0412247	.0130844	3.15	0.002	.0154888	.066960

```

    2013 | .0481795 .0151811 3.17 0.002 .0183194 .078039
> 6
    2014 | .0771433 .0174999 4.41 0.000 .0427223 .111564
> 3
    2015 | .1018548 .0192527 5.29 0.000 .0639862 .139723
> 3
    2016 | .1355829 .0219733 6.17 0.000 .0923631 .178802
> 8
    2017 | .1576611 .0259269 6.08 0.000 .1066647 .208657
> 4
    2018 | .1663453 .0299639 5.55 0.000 .1074085 .225282
> 1
    2019 | .1986627 .0330349 6.01 0.000 .1336855 .263639
> 9
    2020 | .2085136 .035055 5.95 0.000 .139563 .277464
> 2
    2021 | .2315547 .0406217 5.70 0.000 .151655 .311454
> 5
    2022 | .2088143 .0551602 3.79 0.000 .1003183 .317310
> 3
    2023 | .2158077 .0573659 3.76 0.000 .1029734 .328642
> 1
    _cons | 1.737546 .7465081 2.33 0.021 .269221 3.20587
> 1
-----
> -
    sigma_u | .2433518
    sigma_e | .04717234
    rho      | .96378521 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(29, 342) = 248.21 Prob > F = 0.000
> 0

```

203 . est store mechB_s1

204 .

205 . * Step 2: fin_green → carb_dir | ener_rnw

206 . xtreg carb_dir fin_green ener_rnw \$controls i.year, fe

```
Fixed-effects (within) regression      Number of obs   =      36
> 0
Group variable: id                    Number of groups =      3
> 0

R-squared:                            Obs per group:
    Within = 0.2142                      min =      1
> 2
    Between = 0.0203                     avg =     12.
> 0
    Overall = 0.0162                     max =      1
> 2

                                          F(18, 312)      =      4.7
> 2
corr(u_i, Xb) = -0.9702                 Prob > F        =      0.000
> 0
```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_dir						
>]							
> -	fin_green	-.0253773	.0089914	-2.82	0.005	-.0430687	-.007685
> 9							
> 3	ener_rnw	-.2991146	.0983221	-3.04	0.003	-.4925729	-.105656
> 3							
> 1	econ_gdp	.3290901	.0986086	3.34	0.001	.1350682	.523112
> 1							
> 5	econ_pop	-.3370563	.2358239	-1.43	0.154	-.8010626	.1269
> 5							
> 7	econ_fdi	.0087065	.0104611	0.83	0.406	-.0118767	.029289
> 7							
> 9	reg_env	.8169086	.8658691	0.94	0.346	-.8867724	2.5205
> 9							
> 6	reg_str	-.1815838	.4237522	-0.43	0.669	-1.015357	.652189
> 6							
	year						

```

> 2    2013 |   -.0307604   .0226045   -1.36   0.175   -.075237   .013716
> 8    2014 |   -.054831   .0265289   -2.07   0.040   -.1070291  -.002632
> 4    2015 |   -.0434121   .0301319   -1.44   0.151   -.1026996   .015875
> 1    2016 |   -.0498214   .0360575   -1.38   0.168   -.120768   .021125
> 8    2017 |   -.0543909   .0442787   -1.23   0.220   -.1415136   .032731
> 9    2018 |   -.1041229   .0523558   -1.99   0.048   -.2071379  -.001107
> 7    2019 |   -.1219074   .0591985   -2.06   0.040   -.2383861  -.005428
> 5    2020 |   -.1210848   .0636697   -1.90   0.058   -.2463612   .004191
> 1    2021 |   -.1373324   .0755452   -1.82   0.070   -.2859748   .0113
> 7    2022 |   -.2113383   .1023719   -2.06   0.040   -.4127648  -.009911
> 8    2023 |   -.2050877   .1069002   -1.92   0.056   -.4154241   .005248
> 9    _cons |   -.5771758   1.478441   -0.39   0.697   -3.486151   2.33179
> -
      sigma_u |   .16986661
      sigma_e |   .08056418
      rho     |   .8163661   (fraction of variance due to u_i)
> -
F test that all u_i=0: F(29, 312) = 1.41                Prob > F = 0.082
> 9

```

207 . est store mechB_s2

208 .

209 . * Step 3: fin_green → carb_lc | ener_rnw

210 . xtreg carb_lc fin_green ener_rnw \$controls i.year, fe

```
Fixed-effects (within) regression      Number of obs   =      36
> 0                                     Number of groups =      3
Group variable: id
> 0

R-squared:                             Obs per group:
    Within = 0.1039                      min =      1
> 2                                     avg  =     12.
    Between = 0.0443                     max  =      1
> 0                                     Overall = 0.0044
> 2                                     F(18, 312)      =      2.0
> 1                                     Prob > F         =     0.009
corr(u_i, Xb) = -0.9901
```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_lc						
>]							
> -	fin_green	.0407754	.015373	2.65	0.008	.0105276	.071023
> 3	ener_rnw	-.3799196	.1681062	-2.26	0.025	-.7106848	-.049154
> 5	econ_gdp	.2221656	.168596	1.32	0.189	-.1095633	.553894
> 5	econ_pop	-.6350632	.4031998	-1.58	0.116	-1.428398	.158271
> 3	econ_fdi	.027691	.0178858	1.55	0.123	-.0075011	.062883
> 1	reg_env	1.213024	1.480419	0.82	0.413	-1.699843	4.12589
> 2	reg_str	-.1569179	.7245101	-0.22	0.829	-1.582461	1.26862
> 6	year						

```

    2013 | -.0919831 .0386481 -2.38 0.018 -.1680269 -.015939
> 2
    2014 | -.0456899 .0453577 -1.01 0.315 -.1349356 .043555
> 8
    2015 | -.05166 .0515181 -1.00 0.317 -.1530268 .049706
> 7
    2016 | -.0973774 .0616492 -1.58 0.115 -.2186782 .023923
> 4
    2017 | -.0400244 .0757055 -0.53 0.597 -.1889823 .108933
> 6
    2018 | -.1230984 .0895152 -1.38 0.170 -.2992283 .053031
> 5
    2019 | -.1298779 .1012146 -1.28 0.200 -.3290274 .069271
> 5
    2020 | -.0891658 .1088593 -0.82 0.413 -.3033569 .125025
> 4
    2021 | -.1465024 .1291633 -1.13 0.258 -.4006436 .107638
> 9
    2022 | -.2659176 .1750303 -1.52 0.130 -.6103065 .078471
> 3
    2023 | -.2743549 .1827726 -1.50 0.134 -.6339775 .085267
> 7
    _cons | 2.997815 2.527763 1.19 0.237 -1.975803 7.97143
> 3
-----
> -
    sigma_u | .33143614
    sigma_e | .13774455
    rho      | .85271659 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(29, 312) = 0.62 Prob > F = 0.937
> 0

```



```
211 . est store mechB_s3
212 .
213 . esttab mechB_s1 mechB_s2 mechB_s3, ///
> mtitle("fin_green→rnw" "carb_dir(+rnw)" "carb_lc(+rnw)") ///
> keep(fin_green ener_rnw) b(%9.4f) se(%9.4f) star(* 0.1 ** 0.05 *** 0.01
> ) ///
> title("Table 4b. Channel B: renewable electricity share")
```

Table 4b. Channel B: renewable electricity share

	(1) fin_green→rnw	(2) carb_dir(+rnw)	(3) carb_lc(+rnw)
fin_green	0.0161*** (0.0050)	-0.0254*** (0.0090)	0.0408*** (0.0154)
ener_rnw		-0.2991*** (0.0983)	-0.3799** (0.1681)
N	390	360	360

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

```
214 .
215 . * — Channel C: Green dispatch ratio (ratiohour) —
216 .
217 . winsor2 ratiohour, cuts(3 97) replace
218 .
219 . * Step 1: fin_green → ratiohour
220 . xtreg ratiohour fin_green $controls i.year, fe
```

Fixed-effects (within) regression	Number of obs	=	38
> 8			
Group variable: id	Number of groups	=	3
> 0			

```

R-squared:                               Obs per group:
    Within = 0.3482                        min =      1
> 2
    Between = 0.0066                      avg =     12.
> 9
    Overall = 0.0001                      max =      1
> 3

                                           F(18, 340)      =     10.0
> 9
corr(u_i, Xb) = -0.8849                  Prob > F       =     0.000
> 0

```

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	ratiohour						
>]							
> -	fin_green	.0136712	.0042653	3.21	0.001	.0052815	.022060
> 8	econ_gdp	.050214	.0410022	1.22	0.222	-.0304358	.130863
> 9	econ_pop	-.2715536	.1000534	-2.71	0.007	-.4683551	-.07475
> 2	econ_fdi	-.0037047	.0050962	-0.73	0.468	-.0137288	.006319
> 4	reg_env	-1.357631	.4078701	-3.33	0.001	-2.159897	-.55536
> 4	reg_str	-.2815665	.2084237	-1.35	0.178	-.6915287	.128395
> 8							
	year						
	2012	.0403928	.0112166	3.60	0.000	.0183302	.062455
> 5							
	2013	.0282746	.0130194	2.17	0.031	.0026659	.053883
> 2							
	2014	.031882	.0150104	2.12	0.034	.002357	.06140
> 7							
	2015	.0462973	.0165159	2.80	0.005	.0138111	.078783
> 5							
	2016	.0735309	.0188544	3.90	0.000	.0364449	.110616
> 9							
	2017	.0701753	.0222487	3.15	0.002	.0264129	.113937
> 8							
	2018	.059524	.0257188	2.31	0.021	.008936	.11011

```

> 2
    2019 | .0719843 .0283627 2.54 0.012 .0161959 .127772
> 7
    2020 | .0741963 .0300959 2.47 0.014 .0149988 .133393
> 9
    2021 | .0549961 .0350464 1.57 0.118 -.0139388 .123931
> 1
    2022 | .0656679 .0473506 1.39 0.166 -.0274692 .15880
> 5
    2023 | .0468084 .0492465 0.95 0.343 -.0500578 .143674
> 6
    _cons | 2.331116 .6415364 3.63 0.000 1.069236 3.59299
> 6
-----
> -
    sigma_u | .19218938
    sigma_e | .04042691
    rho      | .95762797 (fraction of variance due to u_i)
-----
> -
F test that all u_i=0: F(29, 340) = 34.87          Prob > F = 0.000
> 0

```

```
221 . est store mechC_s1
```

```
222 .
```

```
223 . * Step 2: fin_green → carb_dir | ratiohour
```

```
224 . xtreg carb_dir fin_green ratiohour $controls i.year, fe
```

```

Fixed-effects (within) regression          Number of obs   =       35
> 8
Group variable: id                        Number of groups  =        3
> 0

R-squared:                                Obs per group:
    Within = 0.2570                               min =        1
> 1
    Between = 0.0135                               avg  =       11.
> 9
    Overall = 0.0147                               max  =        1
> 2

```

```

> 6
corr(u_i, Xb) = -0.9756
> 0

```

F(18, 310) = 5.9
Prob > F = 0.000

		Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -	carb_dir						
>]							
> -	fin_green	-.0235336	.0087682	-2.68	0.008	-.0407864	-.006280
> 8	ratiohour	-.564873	.1115972	-5.06	0.000	-.7844567	-.345289
> 3	econ_gdp	.4052699	.0937379	4.32	0.000	.2208269	.589712
> 9	econ_pop	-.5302198	.2315498	-2.29	0.023	-.9858278	-.074611
> 8	econ_fdi	.0070107	.0102624	0.68	0.495	-.0131821	.027203
> 6	reg_env	.06738	.8584845	0.08	0.937	-1.621813	1.75657
> 3	reg_str	-.3951973	.4184971	-0.94	0.346	-1.218651	.428256
> 8							
	year						
	2013	-.039597	.0220209	-1.80	0.073	-.0829264	.003732
> 3	2014	-.070687	.0254773	-2.77	0.006	-.1208174	-.020556
> 6	2015	-.0583863	.0285374	-2.05	0.042	-.1145379	-.002234
> 8	2016	-.0598254	.0337397	-1.77	0.077	-.1262132	.006562
> 5	2017	-.0730581	.0412317	-1.77	0.077	-.1541875	.008071
> 2	2018	-.1313382	.048946	-2.68	0.008	-.2276467	-.035029
> 8	2019	-.151038	.0549594	-2.75	0.006	-.2591786	-.042897
> 3	2020	-.1538069	.0590699	-2.60	0.010	-.2700355	-.037578
> 3	2021	-.1835316	.0703589	-2.61	0.010	-.321973	-.045090
> 2	2022	-.2662383	.09753	-2.73	0.007	-.4581429	-.074333

```

> 8
      2023 |  -0.2740378   0.1018083   -2.69   0.007   -0.4743605   -0.073715
> 2
      _cons |  0.5384073   1.463948     0.37   0.713   -2.342123    3.41893
> 8
-----|-----
> -
      sigma_u |  0.20719617
      sigma_e |  0.07848216
      rho     |  0.87452666   (fraction of variance due to u_i)
-----|-----
> -
F test that all u_i=0: F(29, 310) = 1.84                      Prob > F = 0.006
> 3

```

```
225 . est store mechC_s2
```

```
226 .
```

```
227 . * Step 3: fin_green → carb_lc | ratiohour
```

```
228 . xtreg carb_lc fin_green ratiohour $controls i.year, fe
```

```

Fixed-effects (within) regression      Number of obs   =      35
> 8
Group variable: id                    Number of groups =      3
> 0

R-squared:                            Obs per group:
      Within = 0.1174                      min =      1
> 1
      Between = 0.0371                      avg  =     11.
> 9
      Overall = 0.0036                      max  =      1
> 2

                                           F(18, 310)      =      2.2
> 9
corr(u_i, Xb) = -0.9929                  Prob > F         =      0.002
> 3

```

> -							
>]	carb_lc	Coefficient	Std. err.	t	P> t	[95% conf. interval	
> -							
> 9	fin_green	.0421148	.015299	2.75	0.006	.0120117	.072217
> 4	ratiohour	-.615978	.1947173	-3.16	0.002	-.9991127	-.232843
> 1	econ_gdp	.3192298	.163556	1.95	0.052	-.0025905	.641050
> 4	econ_pop	-.8512965	.4040133	-2.11	0.036	-1.646252	-.056341
> 6	econ_fdi	.0252416	.0179061	1.41	0.160	-.0099913	.060474
> 9	reg_env	.400556	1.497903	0.27	0.789	-2.546787	3.34789
> 6	reg_str	-.3626554	.7302032	-0.50	0.620	-1.799437	1.07412
> 3	year						
> 5	2013	-.1024543	.0384226	-2.67	0.008	-.1780563	-.026852
> 2	2014	-.0657536	.0444534	-1.48	0.140	-.1532223	.02171
> 2	2015	-.0723174	.0497928	-1.45	0.147	-.1702919	.025657
> 5	2016	-.1148759	.0588699	-1.95	0.052	-.2307109	.000959
> 1	2017	-.0686413	.071942	-0.95	0.341	-.2101976	.07291
> 4	2018	-.16205	.0854021	-1.90	0.059	-.3300911	.005991
> 2	2019	-.1730618	.0958944	-1.80	0.072	-.361748	.015624
> 7	2020	-.1372051	.1030665	-1.33	0.184	-.3400034	.065593
> 9	2021	-.212863	.1227638	-1.73	0.084	-.4544188	.028692
> 4	2022	-.3391583	.1701726	-1.99	0.047	-.6739977	-.00431
> 1	2023	-.3633488	.1776374	-2.05	0.042	-.7128763	-.013821
> 4	_cons	4.118425	2.554329	1.61	0.108	-.9075892	9.1444

> -	
sigma_u	.41822274
sigma_e	.13693745
rho	.90317227 (fraction of variance due to u_i)

> 2

230 ■

Table 4c. Channel C: green dispatch ratio

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$